Assignment 3

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Question 1

Suppose all the spins in the lattice are pointing in the same direction (i.e., -1) in the initial configuration. Let the lattice size be $L = 20$. The total magnetic moment (in simulation units) of the entire lattice in this initial configuration will be:

Ans:

$$M = \sum_{i=1}^{N} s_i$$

where $s_i = -1$ for all spins and the total number of spins is

$$N = L \times L = 20 \times 20 = 400.$$

Thus,

$$M = \sum_{i=1}^{400} (-1) = -400.$$

Result:

$$M = -400$$

Question 2

Suppose all the spins in the lattice are pointing in the same direction (i.e., $+1$) in the initial configuration. Let the lattice size be $L = 10$. The total energy (in simulation units where $J_{\text{ising}} = 1$) of the entire lattice will be:

Ans: The Ising Hamiltonian is

$$E = -J \sum_{\langle i,j \rangle} s_i s_j$$

where $s_i = \pm 1$ and the summation is over nearest neighbour pairs.
For the given configuration,

$$s_i = +1 \quad \forall i$$

Thus,

$$s_i s_j = (+1)(+1) = 1$$

The total number of spins is

$$N = L \times L = 10 \times 10 = 100$$

For a triangular lattice, each spin has 6 nearest neighbours. Therefore the total number of bonds is

$$N_{\text{bonds}} = \frac{N \times 6}{2} = \frac{100 \times 6}{2} = 300$$

Hence the total energy is

$$E = -J \times N_{\text{bonds}}$$

Since $J = 1$,

$$E = -300$$

Result:

$$E = -300$$

Question 3

For parameters $k_B T = 4.9$, $L = 10$, and $n_{\text{iter}} = 50000$, the instantaneous magnetization per spin (i.e., the value of magnetic moment per spin in a microstate, M) fluctuates around what value?

Ans: At high temperature ($k_B T = 4.9$), which is possibly above the critical temperature of the triangular lattice Ising model the system is in the paramagnetic phase. In this regime the spins are disordered and the net magnetization averages to zero.
From the simulation,

$$M \approx 0.00173$$

Thus the instantaneous magnetization per spin fluctuates around

$$M \approx 0$$

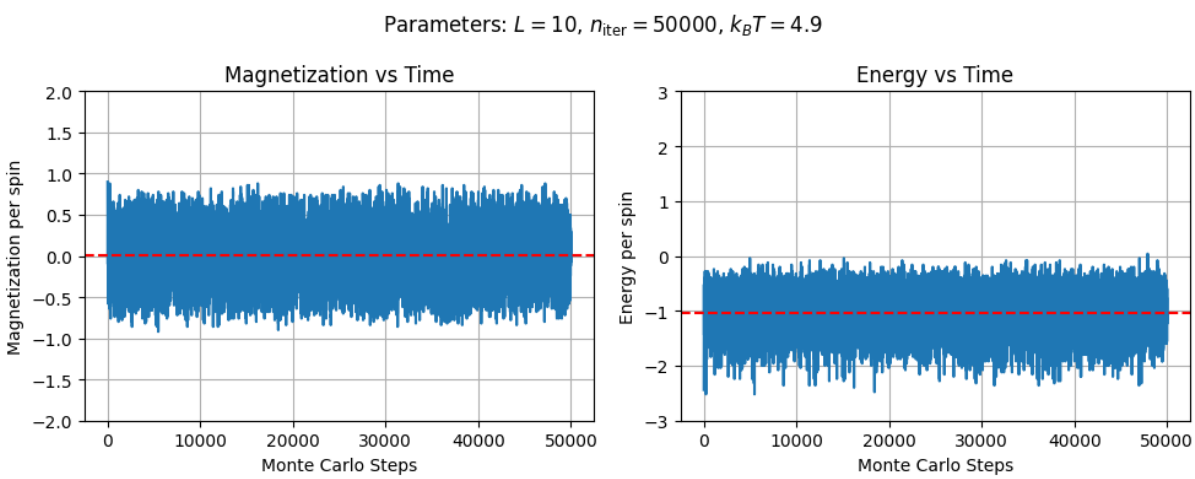


Figure 1: MCS for $k_b T = 4.9$

Initializing all spins to be +1 initially we run 50000 MCS and note the value of the instantaneous magnetization and energy at $k_b T = 4.9, L = 10$. The graph below figure (1) also explain the same.
From the simulation , the measured energy per spin is

$$\frac{E}{N} \approx -1.03.$$

The Hamiltonian of the Ising model is

$$E = -J \sum_{\langle i,j \rangle} s_i s_j,$$

where $s_i = \pm 1$ and the summation is over nearest–neighbour pairs. For a triangular lattice each spin has 6 nearest neighbours, so the energy per spin can be written as

$$\frac{E}{N} = -3J\langle s_i s_j \rangle,$$

where $\langle s_i s_j \rangle$ represents the average nearest–neighbour spin correlation.
The value $E/N \approx -1.03$ therefore implies

$$\langle s_i s_j \rangle \approx 0.34.$$

This shows that although the system is at a relatively high temperature ($k_B T = 4.9$), neighbouring spins still have a positive correlation and tend to align more often than not. This residual short-range ordering arises from the interaction term in the Hamiltonian, which energetically favors parallel spins. As a result, the average energy per spin remains negative rather than approaching zero. The triangular lattice also has a coordination number of six, which increases the number of interacting neighbours and strengthens these local correlations, leading to an equilibrium energy per spin around -1 for the given simulation parameters.

Question 4

For parameters $k_B T = 3.9$, $L = 10$, and $n_{\text{iter}} = 50000$, the instantaneous energy per spin (i.e., the value of the energy per spin in a microstate, E) fluctuates around what value?

Ans For $k_B T = 3.9$, the simulation gives

$$\frac{E}{N} \approx -1.7701$$

and the average magnetization per spin

$$M \approx 0.08675.$$

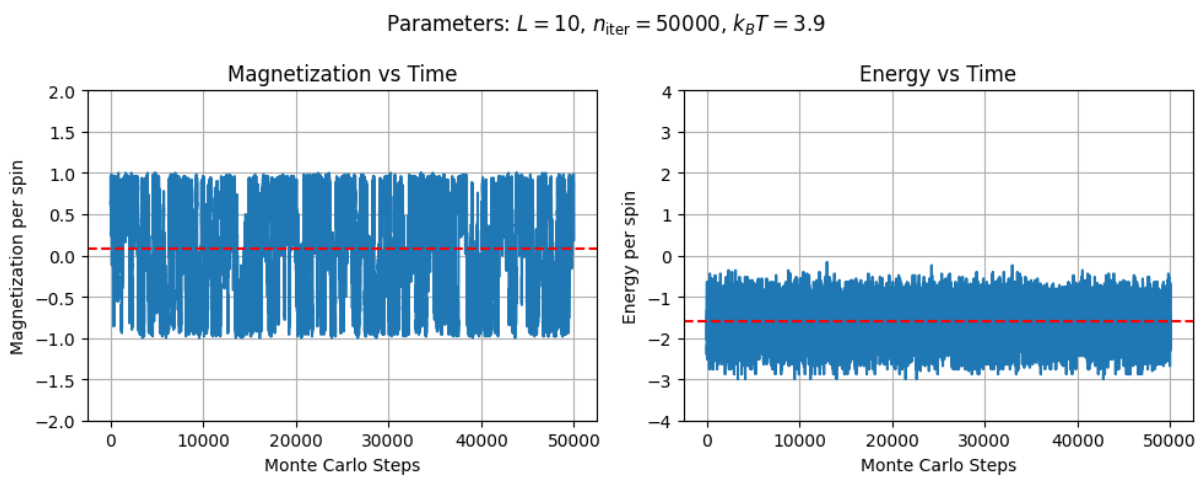


Figure 2: MCS for $k_b T = 3.9$

The Hamiltonian of the Ising model is

$$E = -J \sum_{\langle i,j \rangle} s_i s_j,$$

where $s_i = \pm 1$ and the sum is over nearest neighbour pairs. For a triangular lattice each spin has 6 nearest neighbours, therefore the energy per spin can be written as

$$\frac{E}{N} = -3J\langle s_i s_j \rangle.$$

Substituting the measured value of the energy,

$$\langle s_i s_j \rangle = \frac{1.7701}{3} \approx 0.59.$$

This shows that neighbouring spins are strongly positively correlated and tend to align with one another.
The critical temperature of the triangular lattice Ising model is

$$T_c \approx 3.64.$$

Since $T = 3.9 > T_c$, the system is slightly above the critical temperature and is therefore in the paramagnetic phase. However, because the temperature is close to the critical point, strong short–range correlations persist. As a result, the energy per spin remains significantly negative while the magnetization fluctuates around a small value close to zero.

Question 5

For parameters $k_B T = 4.05$, $L = 10$, and $n_{\text{iter}} = 50000$, the instantaneous energy per spin (i.e., the value of the energy per spin in a microstate, E) fluctuates around what value?

Ans From the simulation with parameters $k_B T = 4.05$, $L = 10$ and $n_{\text{iter}} = 50000$, the measured energy per spin is

$$\frac{E}{N} \approx -1.5923$$

and the average magnetization per spin is

$$M \approx -0.00129.$$

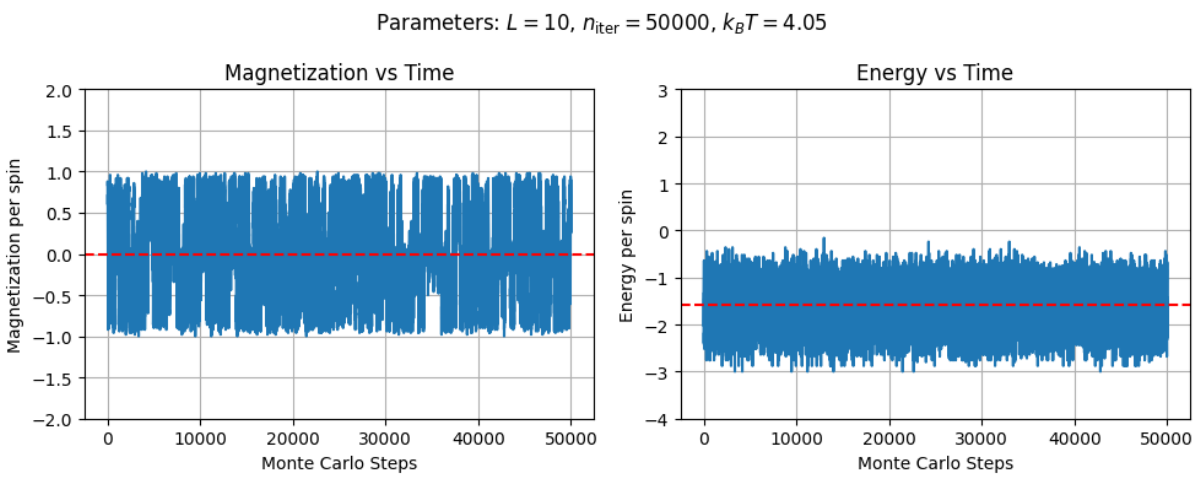


Figure 3: Triangular lattice Ising model with $k_b T = 4.05$

The Hamiltonian of the Ising model is

$$E = -J \sum_{\langle i,j \rangle} s_i s_j$$

where $s_i = \pm 1$ and the summation is over nearest–neighbour pairs. For a triangular lattice each spin has six nearest neighbours, therefore the energy per spin can be written as

$$\frac{E}{N} = -3J \langle s_i s_j \rangle.$$

Substituting the measured value of the energy,

$$\langle s_i s_j \rangle \approx \frac{1.5923}{3} \approx 0.53.$$

This indicates that neighbouring spins still show a positive correlation, although weaker than at lower temperatures.
The critical temperature of the triangular lattice Ising model is approximately

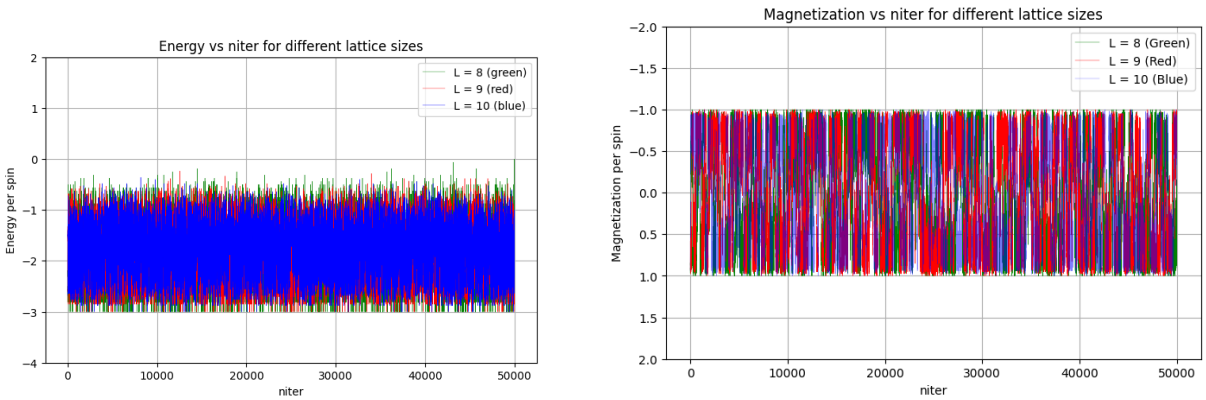
$$T_c \approx 3.64.$$

Since $T = 4.05 > T_c$, the system is in the paramagnetic phase. In this regime the spins are largely disordered, so the magnetization fluctuates around zero, as seen from the simulation value $M \approx -0.00129$. However, because the temperature is not extremely high, short–range correlations remain, giving a negative average energy per spin.

Question 6

Show the fluctuations in the value of magnetization per spin (M) and energy per spin (E) at $T = 3.9$ for lattice sizes $L = 8, 9, 10$ (Fig. 4).

Ans: The fluctuations in the magnetization per spin and energy per spin were obtained from Monte Carlo simulations of the triangular lattice Ising model at $T = 3.9$ for lattice sizes $L = 8, 9$, and 10. The values of M and E were recorded as functions of the number of Monte Carlo iterations.
From Fig. 4 it is observed that both the magnetization and energy exhibit noticeable fluctuations due to thermal effects. However, the magnitude of these fluctuations depends on the lattice size.

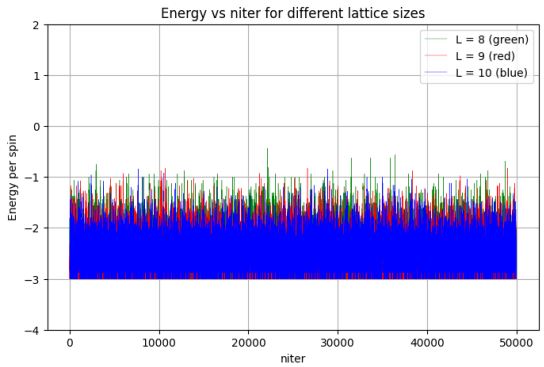


(a) Energy per spin vs iterations

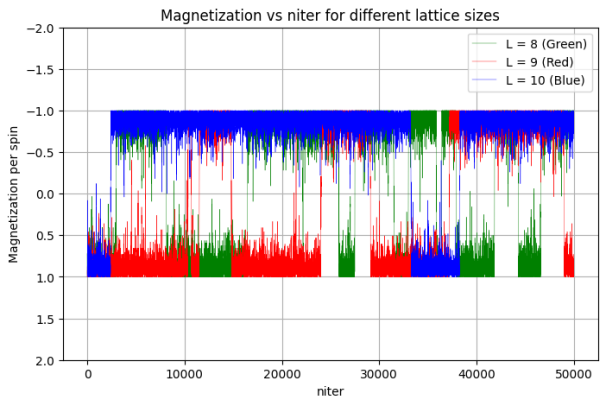
(b) Magnetization per spin vs iterations

Figure 4: Fluctuations in energy and magnetization at $T = 3.9$

For smaller lattices ($L = 8$), the fluctuations are relatively larger because the number of spins in the system is small. In such systems, thermal fluctuations have a stronger effect on the overall magnetization and energy.
As the lattice size increases ($L = 9$ and $L = 10$), the fluctuations become smoother and relatively smaller. This occurs because the system contains a larger number of interacting spins, and the contributions from individual spin flips tend to average out.
Thus, increasing the lattice size reduces the relative magnitude of fluctuations and leads to more stable average values of the magnetization and energy. This behavior is consistent with the general statistical mechanics expectation that fluctuations decrease as the system size increases. Also since we are already operating above the temperature range for directional magnetization ,i.e, paramagnetic regime , we see that the magnetization fluctuate around 0 and the variations are minuscule. Whereas in the case of energy the averaging affect increases for larger lattice size and greater inteactions leads to smoother and less noisy energy variation.



(a) Energy per spin vs iterations



(b) Magnetization per spin vs iterations

Figure 5: Fluctuations in energy and magnetization at $T = 3.3$

In the case of $k_bT = 3.3$ since it is lesser than transition temperature for triangular lattice we get energy averaging around a lesser value , also magnetization remains at a particular state for longer ; residency time increasing as a function of lattice size.

Analysis of the 2D Ising Model at $T = 3.4 < T_c$

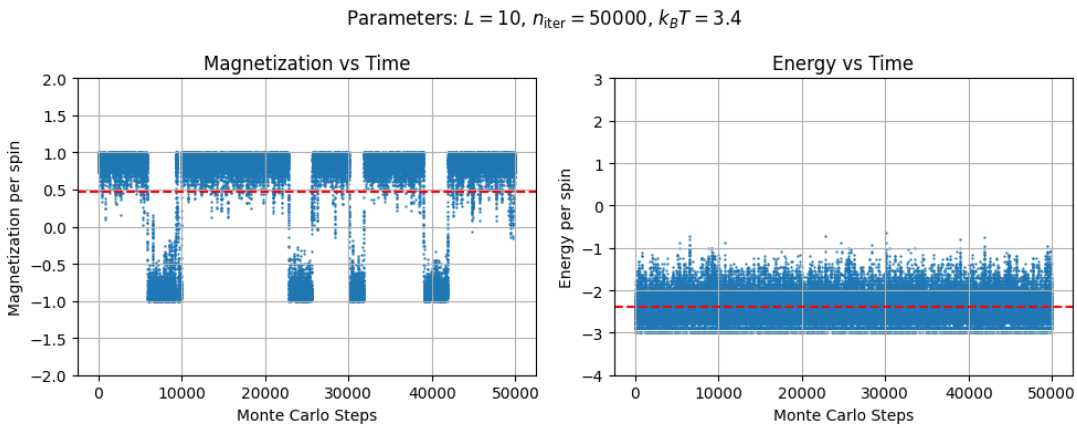


Figure 6: Fluctuations in energy and magnetization at T=3.4

The simulation produced the following results after loading 50,000 Monte Carlo samples:

- Average energy per configuration: $\langle E \rangle = -2.3807$
- Average magnetization: $\langle M \rangle = 0.4728$

For the ferromagnetic Ising model on a triangular lattice with nearest neighbour interactions, the exact critical temperature is given by

$$T_c = \frac{4}{\ln 3} \approx 3.641$$

Since the simulation temperature is

$$T = 3.4 < T_c$$

the system lies slightly below the critical temperature and is therefore expected to remain in the ferromagnetic phase, though close to the critical regime where fluctuations become significant.

Energy Behaviour

The average energy measured in the simulation is

$$\langle E \rangle \approx -2.38$$

In the triangular lattice, each spin interacts with six neighbours, increasing the number of interaction bonds relative to the square lattice case. Consequently, the system has a lower ground state energy per spin due to the larger coordination number.
At temperatures approaching the critical point from below, thermal fluctuations begin to disrupt perfect spin alignment. As a result, the energy increases from its ground state value but remains negative due to the predominance of aligned spin pairs.

Magnetization Behaviour

The measured magnetization is

$$\langle M \rangle \approx 0.473$$

Because the system temperature is below the critical temperature, a non-zero spontaneous magnetization is expected. However, since T is relatively close to T_c, fluctuations reduce the degree of spin ordering. This results in a magnetization value significantly lower than unity, indicating partial alignment of spins across the lattice.

Question 7

Simulations were performed for lattice sizes $L = 26$, $L = 30$, and $L = 34$. The temperature range was chosen as

$$k_B T = 4.0 \text{ to } 3.0$$

with a temperature step of

$$\Delta T = 0.01.$$

At each temperature:

- The system was equilibrated for 10 000 Monte Carlo steps (MCS).
- After equilibration, statistical averages were collected for 50 000 MCS.
- Measurements were taken every Monte Carlo step.

The total number of spins in the lattice is

$$N = L^2.$$

The instantaneous magnetization of the entire lattice is

$$M_L = \sum_{i=1}^{N} s_i,$$

where $s_i = \pm 1$ represents the spin at site i.
The instantaneous energy of the lattice is

$$E_L = -J \sum_{\langle i,j \rangle} s_i s_j,$$

where the sum runs over nearest neighbour pairs.

Magnetization per Spin

The magnetization per spin is defined as

$$m = \frac{M_L}{N}.$$

The average magnetization per spin was computed as

$$\langle m \rangle = \left\langle \frac{M_L}{N} \right\rangle.$$

Observation: At low temperatures the system is ordered and the magnetization per spin is large. As the temperature increases, the magnetization decreases and approaches zero near the critical temperature, indicating a transition to the paramagnetic phase.

Energy per Spin

The energy per spin is defined as

$$e = \frac{E_L}{N}.$$

The average energy per spin was calculated as

$$\langle e \rangle = \left\langle \frac{E_L}{N} \right\rangle.$$

Observation: The energy per spin increases smoothly with temperature. The change in slope near the critical temperature indicates the phase transition.

Magnetic Susceptibility

Magnetic susceptibility was calculated using fluctuations of the total magnetization:

$$\chi = \frac{\langle M_L^2 \rangle - \langle M_L \rangle^2}{k_B T}.$$

Observation: A pronounced peak is observed in the susceptibility near the critical temperature. The height of the peak increases with lattice size, reflecting stronger magnetization fluctuations near the phase transition.

Specific Heat

The specific heat was calculated using energy fluctuations:

$$C_v = \frac{\langle E_L^2 \rangle - \langle E_L \rangle^2}{k_B T^2}.$$

Observation: The specific heat exhibits a peak near the critical temperature. The peak becomes sharper as the lattice size increases, which is characteristic of a second-order phase transition.

Binder Cumulant

To obtain a more accurate estimate of the critical temperature, the Binder cumulant was also calculated. The Binder cumulant is defined in terms of the fluctuations of the total magnetization as

$$U_L = 1 - \frac{\langle M_L^4 \rangle}{3\langle M_L^2 \rangle^2}.$$

Here M_L is the instantaneous magnetization of the entire lattice of size $L \times L$, and the angular brackets denote thermal averages over Monte Carlo configurations.
The Binder cumulant is a dimensionless quantity that is particularly useful for identifying the critical temperature. Unlike quantities such as susceptibility or specific heat, the Binder cumulant has the property that curves corresponding to different lattice sizes intersect at the critical temperature.
Observation: The Binder cumulant curves for different lattice sizes approach each other and intersect near the critical temperature. This crossing point provides a reliable estimate of the phase transition temperature of the system.

Peak Values of Thermodynamic Quantities

From the simulation results for lattice sizes $L = 26$, $L = 30$, and $L = 34$, the peak values of the specific heat and magnetic susceptibility were obtained from the plots. These peaks occur near the critical temperature of the triangular lattice Ising model.

Lattice Size L	$T_{C_v}^{\text{peak}}$	C_v^{peak}	T_χ^{peak}	χ^{peak}
26	3.67	1300.14	3.48	85794.50
30	3.69	1843.91	3.55	134429.42
34	3.68	2376.84	3.56	198773.91

Table 1: Peak values of specific heat and susceptibility obtained from Monte Carlo simulations.

The Binder cumulant curves for different lattice sizes intersect approximately at

$$T_c \approx 3.44.$$

This value provides an estimate of the critical temperature of the triangular lattice Ising model.

Comparison of Thermodynamic Quantities

From the results it is observed that the peak positions of the specific heat C_v are in closer agreement across different lattice sizes than those of the susceptibility χ or Binder cumulant. This behavior can be explained by considering the statistical fluctuations of energy and magnetization.
The specific heat is defined as

$$C_v = \frac{\langle E_L^2 \rangle - \langle E_L \rangle^2}{N k_B T^2},$$

where E_L is the instantaneous energy of the lattice.
Similarly, the magnetic susceptibility is given by

$$\chi = \frac{\langle M_L^2 \rangle - \langle M_L \rangle^2}{N k_B T},$$

where M_L is the total magnetization of the lattice.

Effect of Finite Lattice Size

Energy is a sum over many local spin interactions,

$$E_L = -J \sum_{\langle i,j \rangle} s_i s_j,$$

which averages contributions from all nearest-neighbour pairs in the lattice. As a result, energy fluctuations tend to be smoother and less sensitive to finite-size effects.
In contrast, magnetization is a global quantity,

$$M_L = \sum_i s_i,$$

which depends on the collective alignment of spins across the entire lattice. Near the critical temperature the system frequently flips between positive and negative magnetization states, leading to large fluctuations in M_L. These fluctuations become stronger for smaller lattice sizes.
Consequently, quantities that depend on magnetization fluctuations, such as susceptibility and Binder cumulant,

$$U_L = 1 - \frac{\langle M_L^4 \rangle}{3 \langle M_L^2 \rangle^2},$$

are more sensitive to finite-size effects.

Interpretation

Because the lattice sizes used in the simulation ($L = 26, 30, 34$) are relatively small, finite-size fluctuations affect magnetization more strongly than energy. This leads to greater variability in the susceptibility peaks and Binder cumulant crossings. In contrast, the specific heat shows better agreement across different lattice sizes because energy fluctuations average over many local interactions. As the lattice size increases, these finite-size effects become weaker, and the peaks of all thermodynamic quantities converge toward the true critical temperature of the triangular lattice Ising model,

$$k_B T_c \approx 3.64.$$

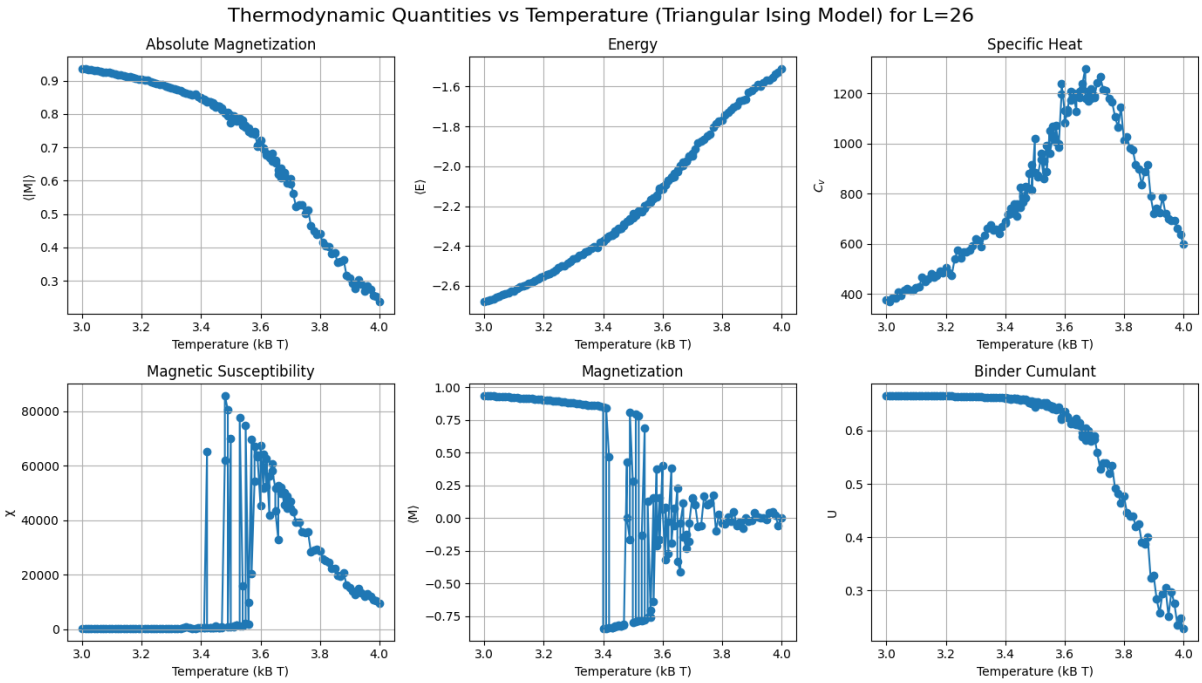


Figure 7: Thermodynamic quantities vs Temperature (L=26)

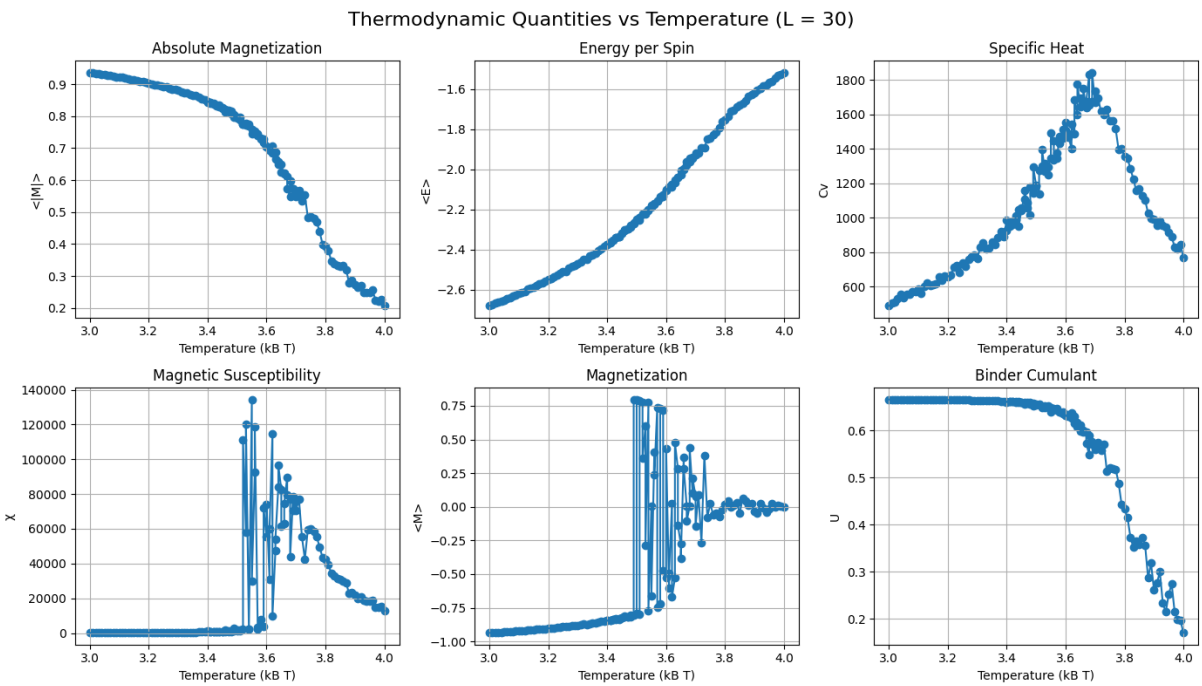


Figure 8: Thermodynamic quantities vs Temperature (L=30)

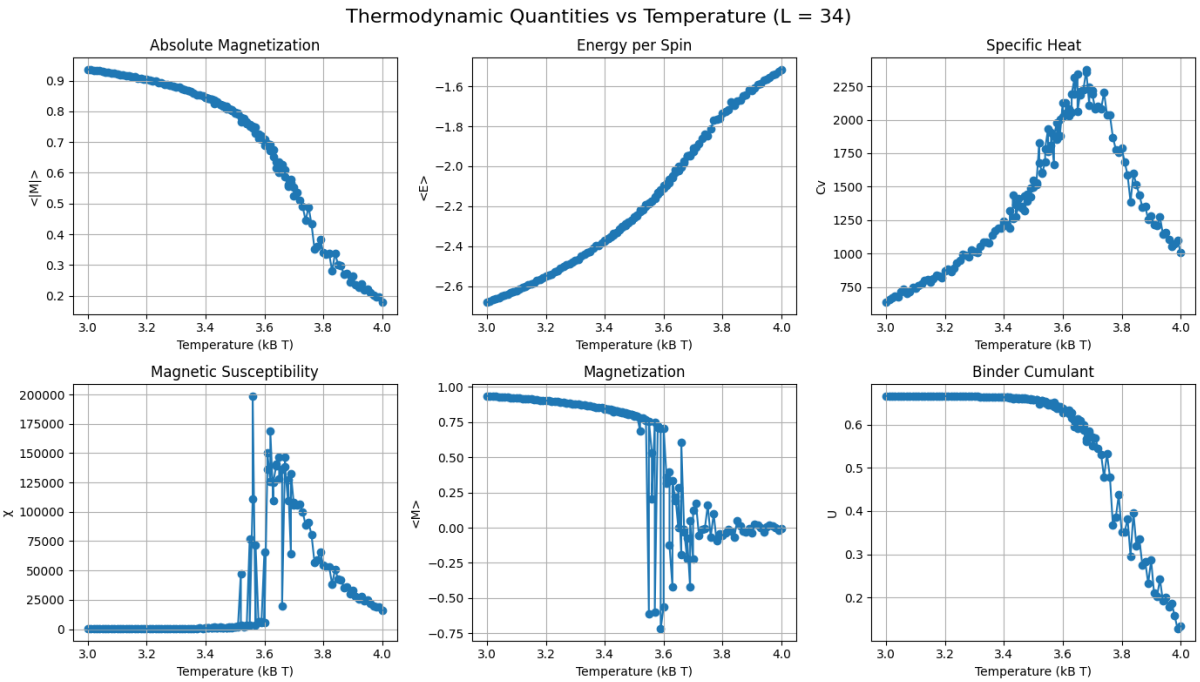


Figure 9: Thermodynamic quantities vs Temperature (L=34)

To find Tc using binder's cumulant:

We find the graph of binder's cumulant U vs Temperature for different lattice sizes 9, 15 and 30 for better visual resolution. The point of intersection gives us the value of T_c as it is a constant irrespective of lattice size. Here $T_c = 3.63$ which is close to the theoretical value for triangular lattice. Here the graphs for L=9,15 are smoother because we average over larger number of iterations.

Also in fig 10 we see that C_v is smoother than M as energy interactions are short ranged and fluctuates slower as compared to χ which depends on spin states of the entire lattice and possess higher long range order.

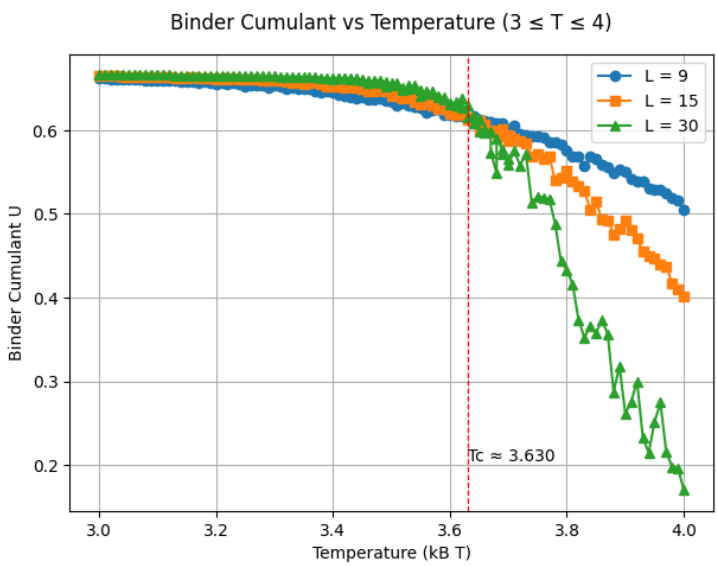
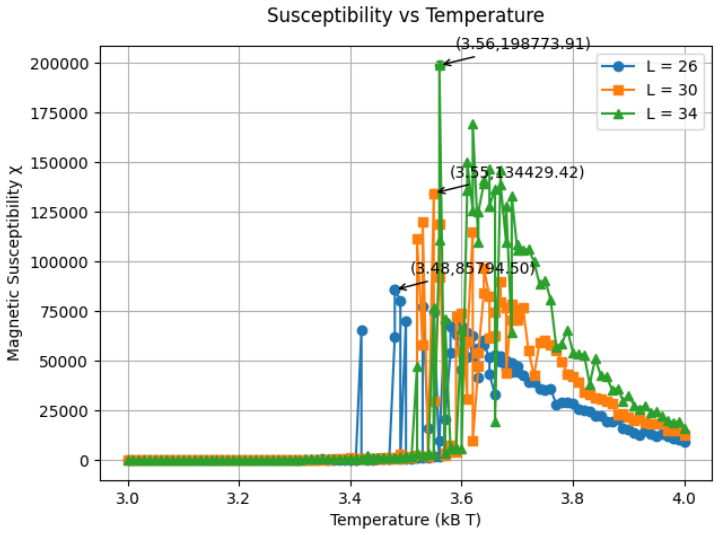
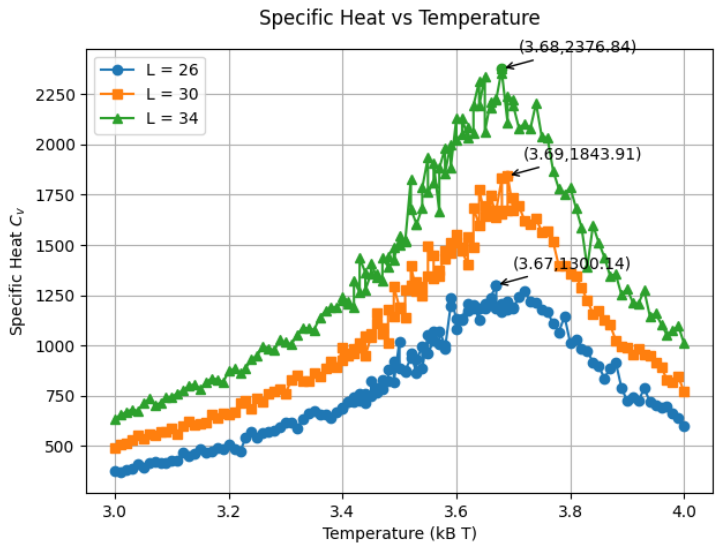


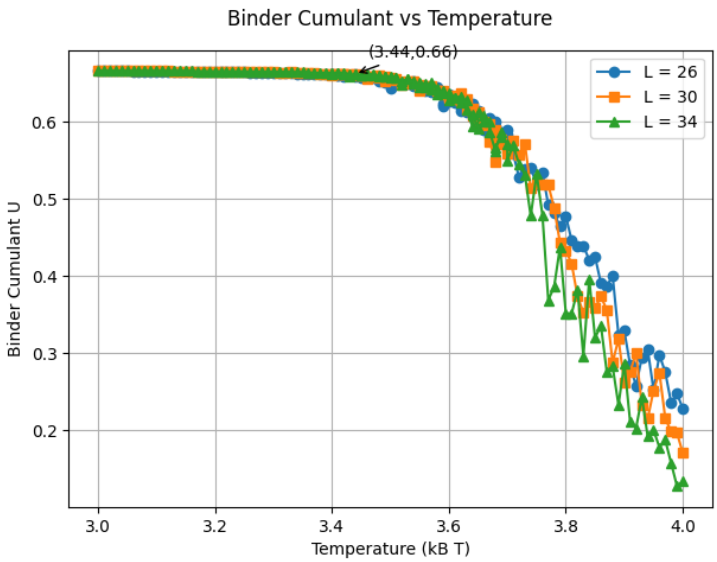
Figure 10: Binder's Cumulant and Tc for L=9,15,30



(a) Chi vs Temp for L=26,30,34



(b) Cv vs Temp for L=26,30,34



(c) Binder's Cumulant vs Temp for L=26,30,34

Figure 11: Comparison of Thermodynamic quantities vs Temp across lattice sizes

Question 8

There are multiple energy levels $(E_1, E_2, \ldots, E_n, \ldots)$ in a system in equilibrium at temperature T. The average number of particles in energy level E_5 is 100, and the average number of particles in energy level E_{10} is 50. The energy at level E_{10} is greater than that at level E_5. The number of particles jumping from E_5 to E_{10} is 10 per second. Determine the number of particles jumping per second from E_{10} to E_5.

Ans: At thermal equilibrium, the principle of *detailed balance* holds. This means that the rate of transitions from one energy level to another must equal the reverse transition rate.
Let

$$N_5 = 100, \qquad N_{10} = 50$$

be the average populations of the two levels.
Let

$$W_{5\to 10}$$

be the transition probability per particle from E_5 to E_{10}, and

$$W_{10\to 5}$$

be the transition probability per particle from E_{10} to E_5.
The transition rates are

$$R_{5\to 10} = N_5 W_{5\to 10}$$

$$R_{10\to 5} = N_{10} W_{10\to 5}$$

Since the system is in equilibrium,

$$R_{5\to 10} = R_{10\to 5}$$

We are given

$$R_{5\to 10} = 10 \text{ particles per second.}$$

Therefore,

$$R_{10\to 5} = 10 \text{ particles per second.}$$

$$\boxed{10 \text{ particles per second}}$$